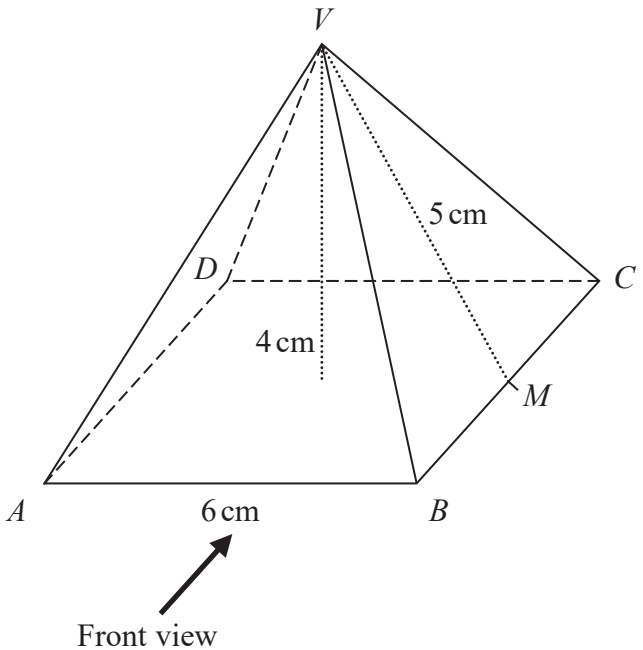
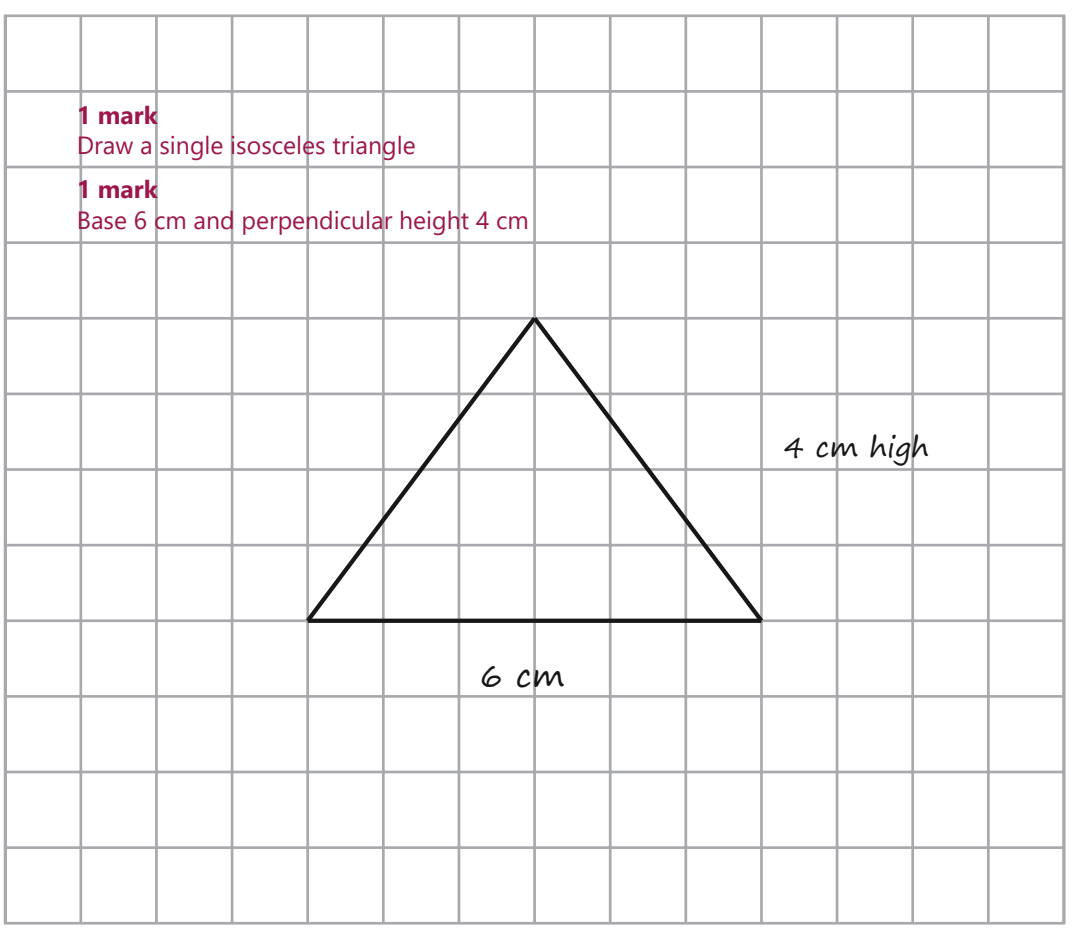
23 Here is a solid square-based pyramid, *VABCD*.



The base of the pyramid is a square of side 6 cm.
The height of the pyramid is 4 cm.
M is the midpoint of *BC* and *VM* = 5 cm.

(a) Draw an accurate front elevation of the pyramid from the direction of the arrow.



(2)

(b) Work out the total surface area of the pyramid.

One triangular face:

$\frac{1}{2} \times 6 \times 5 = 15 \text{ cm}^2$

1 mark
Use the sloping height 5 cm for a triangular face

Four faces: $4 \times 15 = 60 \text{ cm}^2$

Square base: $6 \times 6 = 36 \text{ cm}^2$

1 mark
Add the four triangles and the base

Total surface area = $60 + 36 = 96 \text{ cm}^2$

1 mark
Correct numerical area

1 mark
Use square centimetres

96 cm^2

(4)

(Total for Question 23 is 6 marks)